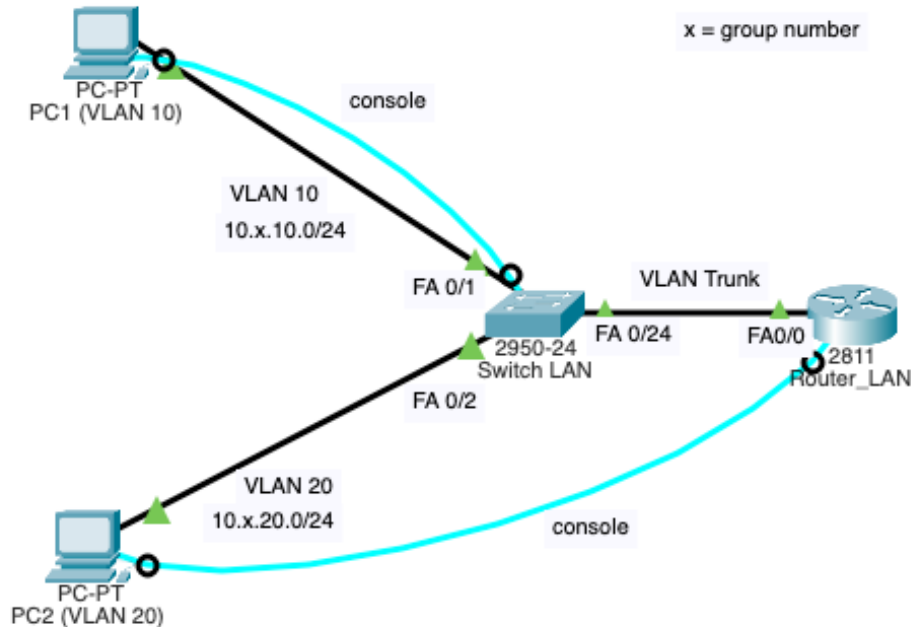


Hardware Lab: VLAN and DHCP configuration



Preparations

Before starting, the lab administrator will assign you a group number and a set of hardware components to your group. Each set comprises a hub, a switch, and three routers. In this lab, you'll only need to use one router, which you can choose freely.

Note: Currently, the Hochschule Bremen networking lab can support up to seven hardware groups.

To begin, discuss the tasks within your group and assign the two lab PCs as either PC1 or PC2. Then, prepare the lab PCs by accessing "VMWare Workstation" using the desktop shortcut and start Windows10 as a virtual machine on each PC.

After starting the VM, access the properties of the network connections. Each VM has an internet and a lab LAN connection. Remember that only one of these connections can be active at the same time. To use the lab hardware, activate the lab LAN connection and disable the internet connection. However, keep in mind that the lab LAN connection may still require IP address configuration after activation.

1. Cable your topology. For cabling the PCs, use the patch port information provided for every PC. Connect the Ethernet ports of the PCs to the Cisco Switch of your hardware group. Additionally connect the console ports of a PC to the console ports of the router and another PC to the console port of the switch. Unlike with packet tracer the router / switch needs a physical connection to their console ports to access the CLI. In order to access the CLI of a router/switch over the console port of the PC you need to start a terminal program like Hyper Terminal or Tera Term. Start the terminal program and use the COM1 port of the PC for this connection. Configure the terminal session with the following parameters:

Terminal Configuration [X]

Port Configuration

Bits Per Second: 9600

Data Bits: 8

Parity: None

Stop Bits: 1

Flow Control: None

OK

Note: After the terminal session is correctly configured and the router/switch have booted successfully, press the <enter> key to access the prompt. Whenever there is no startup configuration present on the router you will be asked to use a setup dialog:

<Continue with configuration dialog? [yes/no]: >

To start with the manual configuration of the router, quit the configuration dialog with <n> or use the break command <control+c>.

2. Access the CLI of the switch and the router and display the running configuration. There should not be any configuration present on the switch as well as on the interface FastEthernet 0/0 of the Router_LAN.

VLAN Configuration

1. The first step will be to configure the switch. The two PCs will become members of two different VLANs.

Create VLANs 10 and 20 on the switch.

Configure Switchport FastEthernet 0/1 to access VLAN10.

Configure Switchport FastEthernet 0/2 to access VLAN20.

Configure Switchport FastEthernet 0/24 as a trunk port.

2. Configure the Router_LAN:
 Activate the interface FastEthernet 0/0 (do not configure an IP address)
 Create a subinterface FastEthernet 0/0.10, choose VLAN10, dot1Q as encapsulation and the ip address 10.x.10.1 /24 and activate this interface.
 Create a subinterface FastEthernet 0/0.20, choose VLAN20, dot1Q as encapsulation and the ip address 10.x.20.1 /24 and activate this interface.
3. Configure PC1 with an IP address of 10.x.10.11 from VLAN10 and configure PC2 with an IP address 10.x.20.11 from VLAN20. What gateway information will be used for both PCs? What is the netmask corresponding to /24?

From PC1, try to ping PC2. The ping should be successful.

DHCP Configuration

Configure the Router_LAN:

1. Configure a DHCP pool with the name of "vlan10" to use network 10.x.10.0/24 and default router 10.x.10.1

Configure a DHCP pool with the name of "vlan20" to use network 10.x.20.0/24 and default router 10.x.20.1

2. Prevent the following addresses to be used by DHCP:

10.x.10.1 – 10.x.10.10

10.x.20.1 – 10.x.20.10

3. Set the PCs to automatically obtain IP addresses and check the configuration by using the "ipconfig" command and pinging from PC1 to PC2

Documentation for further use

1. Save the configuration of the switch in a text file named switch_lan.txt
2. Save the configuration of the router in a text file named router_lan.txt

Keep the two files handy for the next lab.